Minimum shaft diameter calculation

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1. Scope

This document provides a comprehensive guide for the calculation module **Mod.1130** "Minimum shaft diameter calculation". The purpose of **Mod.1130** is to determine the **MND** of pump shafts to ensure adequate mechanical strength under operating and test conditions. It performs the following key functions:

- Calculation of the **MND**.
- Verification of the suitability of the job motor for test conditions.

The procedure is applicable to the plant of Nova Milanese and the plant of La Spezia.

2. Definitions

BBx Between Bearings (type of pumps)

DOL Direct On Line

MND Minimum Nominal Diameter of shaft

SF Safety Factor

STC Starting Torque Curve

TC Test Center

VS Vertically Suspended (type of pumps)

3. User instructions and calculation notes

Mod.1130 is organized into multiple sheets:

- **1 - Input data and calculation:** this sheet is designated for entering all input data, and calculations are performed here accordingly.
- **2 - Material Database-INFO ONLY**: this sheet provides a reference database for material properties, including data in function of temperature. It is intended for information only.

For calculation instructions and for more information about shaft materials, see paragraph 3.1.

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3.1. Mod.1130 user instructions

To perform the **MND** calculation, the user must first complete all required input fields in the **"1 - Input data and calculation"** sheet. Once the inputs are entered, the user should press the designated button shown in *Fig. 1*. The calculation is considered complete when the message **"Solution is updated"** appears in the green notification box. If any input value is manually modified afterward, the button status will change to **"Click to update"** (displayed in a red box), indicating that the calculation needs to be refreshed.

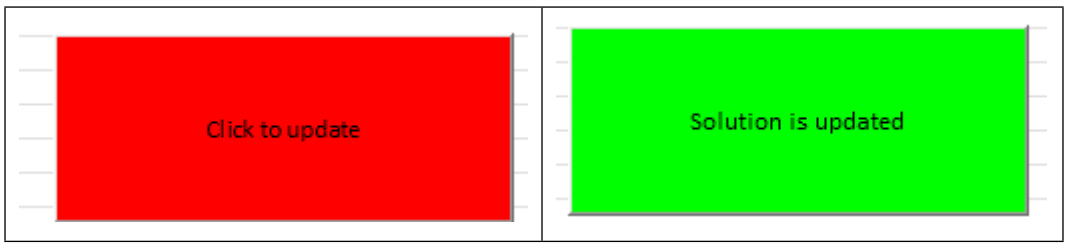

Fig. 1: Calculation button.

- Section "**JOB AND ITEM DATA**": Complete the required basic data fields.
- Section "**PUMP CONSTRUCTION DATA**": Complete the required data fields for the selected item using the annotations provided within the calculation sheet as guidance. Notes:
  - While the sheet titled "**2 - Material Database-INFO ONLY**" serves as a helpful reference for material data, the governing sources for material specifications are **ASME BPVC Section II, Part D** and the relevant **ASTM standards** (i.e., ASTM A276). A list of ASTM standards involved in the most common shaft materials are listed in Chapter 4 ("References"). These standards must be consulted to confirm material properties, including the required bar condition (e.g. annealed, hot-finished, cold-finished, etc.). For user convenience, the **"2 - Material Database – INFO ONLY"** sheet provides clarification notes and references to relevant ASTM standards and bar conditions, helping users accurately interpret the data.
  - In this section, it is also checked whether the job's motor is suitable for the test conditions (*Fig. 2*). The calculation module automatically issues a warning if any incompatibility is detected. If the job's motor is deemed unsuitable, follow these instructions:

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- Ask **TC** to confirm the nominal power of the motor intended for testing the actual item. This power must be no less than the value specified under 'Pump maximum absorbed power at test [kW]'.
- Enter the nominal power provided by **TC** for the test motor in the field labelled 'Nominal Motor Power [kW]'.
- Set the 'Starting Torque Coefficient' field to 1.9. This value is recommended because the starting process in the **TC** is typically very smooth. For more information about calculation coefficients see further below.
- Run the calculation using the provided button (*Fig. 1*) and review the resulting output.

Pump maximum absorbed power at test [kW]	3679
Notes about driver	Job motor cannot be used for test. Verify shaft using Test Center motor power

Fig. 2 – Job's motor incompatibility, check for test conditions.

- Section "**MINIMUM NOMINAL DIAMETER CALCULATION PARAMETERS**": Complete the required data fields for the selected item, using the annotations provided within the calculation sheet as guidance. Notes:
  - '**α1 - Notch Factor'**: it accounts for the presence of notches (typically keyways) on the shaft. Values less than 1 increase the calculation **SF**. Values for this factor are based on internal know-how:
    - α_1 = 1: keyway is not foreseen. No detrimental effects from notches.
    - α_1 = 0.8: keyway is foreseen - other pumps apart from **BB3/BB4/BB5** pumps.
    - α_1 = 0.625: keyway is foreseen - **BB3/BB4/BB5** pumps.
  - '**α_2 - Discontinuity Torque Factor**': it accounts for eventual discontinuity in torque coming from the driver. Values less than 1 increase the calculation **SF**. Values for this factor are based on internal know-how:
    - α_2 = 1: continuous torque from electric motors or steam turbines. No detrimental effects from torque fluctuations.
    - α_2 = 0.7: discontinuous torque from diesel engines.
  - '**Starting Torque Coefficient**': it is a calculation coefficient defined as the ratio of the initial torque delivered by a driver at startup to its rated steady-state

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torque. Values greater than 1 increase the calculation **SF**. These values shall be considered:

- 2.5 for **DOL** motor startup or for **BB3/BB4/BB5** pumps (including soft starters for these pumps).
- 1.9 for soft starters, except for **BB3/BB4/BB5** pumps (see above).
- 1.9 for soft startup at **TC** including **BB3/BB4/BB5**.
- If driver data are available from **STC**, the datum of the startup multiplier with respect to the rated torque can be considered, and the above proposed values can be disregarded.

- ‘**Material Safety Factor**’: it is an additional **SF** related to the shaft material. It also accounts for the use of materials with relatively low mechanical strength. Values greater than 1 increase the calculation **SF**. These values shall be considered:
  - 1.2 for shaft structural steels other than 304/316 stainless steels or for **BB3/BB4/BB5** pumps, independently from shaft material.
  - 1.4 for shaft structural stainless steels like 304/316 except for **BB3/BB4/BB5** pumps (see above). However, by applying this safety factor to **BB3/BB4/BB5** pumps, the calculation results are more conservative.

- Section “**MINIMUM NOMINAL DIAMETER CALCULATION OUTPUT**”: **MND** calculation output in [mm].

3.2. Mod.1130 calculation notes (formulas)

The purpose of these calculation notes is to illustrate the derivation of the formulas used in the **Mod.1130**. Specifically, to determine the **MND** of a shaft subjected to torsion, the following formula is derived from the fundamental torsion equation:

$$MND = \left(\frac{16 \cdot M_t}{\pi \cdot \tau_{max}}\right)^{1/3}$$

Where:

- $\mathbf{M_t}$ is the applied torque on shaft.
- $\boldsymbol{\tau_{max}}$ is the maximum allowable shear stress for the shaft material.

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By substituting the expression for τ_{max} in terms of bending stress, and considering that the maximum stress must not exceed the allowable stress of the shaft material (accounting for the appropriate safety factor), it can be reformulated as follows:

$$MND = \left(\frac{16 \cdot M_t}{\pi \cdot \frac{s_{y_shaft}}{\sqrt{3} \cdot SF}} \right)^{1/3}$$

Where:

- **S_{y_shaft}** is the shaft material yield stress evaluated at design temperature.
- **SF** is the global safety factor with respect to the yield stress.

When designing a shaft, it is essential to consider various factors that influence its strength. These factors are collectively included in the global **SF**. The calculation formula can thus be expressed also as:

$$MND = \left(\frac{16 \cdot M_t}{\pi \cdot \frac{s_{y_shaft} \cdot \alpha_1 \cdot \alpha_2}{\sqrt{3} \cdot K}} \right)^{1/3}$$

Where:

- **α_1** is the “Notch Factor” (refer to paragraph 3.1 for more information).
- **α_2** is the “Discontinuity Torque Factor” (refer to paragraph 3.1 for more information).
- **K** represents the portion of the global **SF** derived from the product of the ‘Starting Torque Coefficient’ and the ‘Material Safety Factor’ (refer to paragraph 3.1 for more information).

When the axial thrust gives not negligible axial stresses on the shaft, such as in multistage **VS** pumps with small shaft diameters, the **MND** must be determined using the following equation. Since the equation is implicit with respect to **MND**, in **Mod.1130** the value is obtained through an iterative process, facilitated by a macro embedded in the calculation sheet.

$$S_{y_shaft} = \frac{K}{\alpha_1 \cdot \alpha_2} \cdot \sqrt{3 \cdot \left[\frac{M_t \cdot MND}{2 \cdot \left[\frac{\pi}{2} \cdot \left[\left(\frac{MND}{2} \right)^4 \right] \right]} \right]^2 + \left[\frac{F_{axial}}{\pi \cdot \left(\frac{MND}{2} \right)^2} \right]^2}$$

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Where:

- $\mathbf{F_{axial}}$ is the pump axial thrust.

4. References

- **Mod.1130**
- API 610
- ASME BPVC, Section II, Part D
- ASTM A276
- ASTM A29
- ASTM A434
- ASTM A564
- ASTM B865
- ASTM B446

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